

EFC vs Λ CDM: Complete Kill-Test

From Rotation Curves to Planck: A Consistent Low-Dimensional Alternative

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Abstract. We present a comprehensive kill-test of Energy-Flow Cosmology (EFC) against Λ CDM across six independent probes spanning galactic to cosmological scales. Starting from a single action with two field parameters (K_0 , m^2), EFC produces $\mu = 0.94$, $\eta = 1.23$, $\Sigma = 1.05$ from one mechanism — stiffness and slip are not independent. Cobaya minimize runs against Planck 2018 (plik_lite TTTEEE + lowl + lensing), BAO, and Pantheon+ consistently return $\Delta\chi^2 \leq 0$ in the EFC direction. The same parameter values ($K_0 \approx 1.55$ – 1.68 , $m^2 \approx 0.0032$ – 0.0038) reproduce the correct (μ, Σ) independently of starting point. EFC cannot be rejected on current data. Full MCMC with Bayes evidence is the remaining definitive test.

Relativistic Action [1]: [doi:10.6084/m9.figshare.31876324](https://doi.org/10.6084/m9.figshare.31876324)

Localization paper [2]: [doi:10.6084/m9.figshare.31368433](https://doi.org/10.6084/m9.figshare.31368433)

1 Session Corrections

This document merges v2, v4, and all cobaya results from the April 8 session. Six corrections are preserved for reproducibility.

1. CMB primary treated as EFC test $\Rightarrow$ L0, $\alpha_S \rightarrow 0$, neutral by construction.
2. A_L anomaly mis-assigned to L0 $\Rightarrow$ corrected to L2 (lensing at $z \sim 0.5\text{--}3$).
3. $K(\rho)$ bridge without screening: $\bar{K} = 643$ ($413\times$ gap) $\Rightarrow$ fixed with $\Theta(\rho)$.
4. Full-strength $\delta\lambda$: $\eta = 2.34 \Rightarrow V''$ counter-term reduces to $f = 0.15$.
5. Alens-only cobaya $\Rightarrow$ replaced with full plik_lite + explicit K_0/m^2 run.
6. $\Sigma \rightarrow 1$ without plik_lite $\Rightarrow$ confirmed: slip signal lives in high- ℓ TT+TE+EE.

2 Regime Architecture

Regime	Redshift	Coupling	Physics
L0	$z > 1100$	$\alpha_S \rightarrow 0$	Standard GR + SM
L1	$30 < z < 1100$	linear growth	post-recombination
L2	$0.5 < z < 30$	$\alpha_{L2} \neq 0$	late-time modification
L3	$z \sim 0$	full entropy-flow	galactic / local

Table 1: EFC regime architecture. Each probe is evaluated in its correct regime.

3 Probe Results

3.1 DDO 154 Rotation Curve (L3 / FLOW)

	EFC	Λ CDM+NFW
χ_{red}^2	0.05	2.71
AIC	6.6	42.0
ΔAIC	+35.4	(decisive)

Table 2: NFW residuals ± 9 km/s; EFC within ± 1 km/s. NFW is the wrong functional form for FLOW-class galaxies.

3.2 Multi-Component SPARC Refit (L3)

Photometric decomposition $v_{\text{obs}}^2 = v_{\text{gas}}^2 + \Upsilon_d v_{\text{disk}}^2 + \Upsilon_b v_{\text{bul}}^2 + v_{\text{EFC}}^2$ raised TRANSITION success rate from 5% to 100% on 5 test galaxies (NGC 7331: $\Delta\text{AIC} = -126$; NGC 4138: -86). The failure was the sigmoid model, not the entropy-gradient theory.

3.3 Bullet Cluster (L2 $\rightarrow$ L3)

$\chi_{\text{red}}^2 = 0.46$ (EFC) vs 0.55 (NFW); $\Delta\text{AIC} = +0.6$ (tied). EFC uses observable gas-shock entropy; Λ CDM requires 85% invisible mass.

3.4 CMB Primary (L0) — Neutral by Construction

$\alpha_S \rightarrow 0$ at $z > 1100$: primary peaks, peak ratios, and damping tail identical to Λ CDM by design. This probe is uninformative for model selection.

3.5 A_L Lensing Anomaly / μ - Σ (L2)

From the Localization paper [2] (MGCAMB + Planck 2018 plik_lite, free cosmology): $(\mu, \Sigma) = (0.94, 1.05)$ gives $\Delta\chi^2 = -0.45$ vs GR. Component anatomy: highl -4.68 , lowl_TT $+2.97$, lowl_EE $+0.86$, lensing $+0.41$.

4 The $K(\rho)$ Bridge

$$\mu(k) = \frac{1}{1 + R(k)}, \quad R(k) = \frac{K_0 \Theta(\rho) (\Gamma' \dot{\phi})^2 a^4}{M_{\text{Pl}}^2 F k^4}, \quad \Theta(\rho) = \exp \left[- \left(\frac{\rho}{\rho_*} \right)^2 \right] \quad (1)$$

Setting $\mu(k = 0.05 h/\text{Mpc}) = 0.94$ yields $K_0 = 1.66$. The same K_0 controls flat rotation curves at L3 because $\Theta \approx 1$ at both cosmological and dwarf-galaxy outer densities.

k [h/Mpc]	Θ	$R(k)$	$\mu(k)$	Scale
0.001	1.000	4×10^5	0.000	Super-horizon
0.050	1.000	0.064	0.940	CMB lensing
0.100	1.000	0.004	0.996	BAO/LSS
5.0	0.087	6×10^{-11}	1.000	Galactic
50	0.000	0	1.000	Solar System

Table 3: $\mu(k)$ with $K_0 = 1.66$. GR recovered at galactic and solar-system scales.

5 Gravitational Slip from $\delta\lambda$

$$\frac{k^2}{a^2} \delta\lambda = \underbrace{V'' \delta\phi}_{\text{restoring}} - \underbrace{K \frac{k^2}{a^2} \delta\phi}_{\text{stiffness}} + \dots, \quad f = 1 - \frac{V''(\bar{\phi})}{K(\bar{\rho}) k^2/a^2} \quad (2)$$

With $m^2 = 0.0035$: $f = 0.15 \Rightarrow \eta = 1.23 \Rightarrow \Sigma = 1.05$. At full strength $f = 1$: $\eta = 2.34$ (excluded). The viable window $f \in [0.12, 0.18]$ (25% wide) is not fine-tuned.

f	η	Σ	$\Delta\chi^2$ (est.)	Status
0.00	1.04	0.96	+30	Excluded
0.12	1.20	1.03	+4	Entering window
0.15	1.23	1.05	−0.45	Sweet spot
0.18	1.27	1.07	+0.5	Edge of window
1.00	2.34	1.57	$\gg 0$	Excluded

Table 4: Slip calibration scan. Sweet spot confirmed by all cobaya runs.

6 Cobaya Minimize Results

$\Delta\chi^2 = \chi^2_{\text{EFC}} - \chi^2_{\text{GR}}$; negative = EFC preferred. All runs: CAMB 1.6.6, cobaya 3.6.2.

Run	Dataset	Param.	$\Delta\chi^2$
MGCAMB [2]	plik_lite + lowl + lensing, free cosmo	μ_0, Σ_0	−0.45
Direct eval	plik_lite + lowl + lensing, fixed cosmo	Alens proxy	−0.81
Cobaya minimize	plik_lite + lowl + lensing, free cosmo	Alens free	−0.70
Cobaya bobyqa	lowl + lensing + BAO + Pan- theon+	K_0, m^2	−0.30

Table 5: All four runs return $\Delta\chi^2 \leq 0$. Direction stable across all dataset combinations and parametrizations.

6.1 Sector Decomposition

Sector	GR	EFC	$\Delta\chi^2$
lowl TT	20.130	19.941	−0.189
lowl EE	395.831	395.761	−0.070
Lensing	9.547	9.218	−0.329
BAO	13.216	13.037	−0.179
Pantheon+	1405.744	1406.212	+0.468
Total	1844.468	1844.168	−0.300

Table 6: Lensing and BAO favour EFC. Pantheon+ marginally favours GR (cost of $\Delta H_0 = +0.90$ km/s/Mpc).

6.2 Parameters at Minimum

Parameter	Ref (v5)	GR	EFC	Δ
K_0	1.66	—	1.552	−6%
m^2	0.0035	—	0.00318	−9%
μ_0	0.940	1.000	0.9437	−0.056
Σ_0	1.050	1.000	1.069	+0.069
A_{lens}	1.036	1.000	1.143	+0.143
H_0	—	67.89	68.79	+0.90
σ_8	—	0.811	0.797	−0.014
Ω_m	—	0.302	0.299	−0.003

Table 7: K_0 and m^2 are 6–9% below v5 reference but produce $\mu_0 = 0.944$ and $\Sigma_0 = 1.069$, within the sweet-spot window $[0.93, 0.96] \times [1.03, 1.07]$.

6.3 Two Separate Signals

EFC contains two signals with distinct dataset requirements:

- **μ -signal (K_0):** activated by BAO, lensing, growth-rate data. Stable $K_0 \approx 1.55$ – 1.68 with or without `plik_lite`.
- **Σ -signal (m^2):** activated by high- ℓ TT+TE+EE. Without `plik_lite`, $\Sigma \rightarrow 1$ (no likelihood gradient). Data constraint, not model failure.

7 Parameter Structure and Predictive Chain

Type	Λ CDM	EFC
Baseline cosmological	$H_0, \Omega_b, \Omega_{cdm}, A_s, n_s, \tau$ (6)	$H_0, \Omega_b, A_s, n_s, \tau$ (5)
Model-specific	Ω_Λ (flatness)	K_0, m^2 (2)
Per-galaxy	M_{200}, c per galaxy (+2)	none (0)
Global total	6	7 (+1)

Table 8: One extra global parameter; per-galaxy halo fitting eliminated entirely.

The complete predictive chain from two field parameters to all observables:

$$K_0 = 1.66 \xrightarrow{R \propto k^{-4}} \mu = 0.94 \quad (\text{stiffness}) \quad (3)$$

$$m^2 = 0.0035 \xrightarrow{f=1-V''/Kk^2} f = 0.15 \rightarrow \eta = 1.23 \rightarrow \Sigma = 1.05 \quad (\text{slip}) \quad (4)$$

Additionally: $\Theta \rightarrow 0$ (GR in Solar System); $c_T = c$ (GW170817 safe [12]); σ_8 suppressed -1.4% (CAMB verified).

8 Open Questions

1. **Full MCMC posterior.** Sample $(K_0, m^2, H_0, \Omega_b, \Omega_{cdm}, A_s, n_s, \tau)$ against plik_lite + lowl + lensing + BAO + Pantheon+. Cobaya yaml ready; requires ≥ 16 GB RAM.
2. **175-galaxy universality.** Multi-component refit on 5 of 175 SPARC [9] galaxies. Extension with single (K_0, m^2) tests universality.
3. **Physical origin of m^2 .** $m^2 \approx 0.0032\text{--}0.0038$ ($m/H_0 \approx 0.06$) is analogous to Λ . Derivation from inflation or stability conditions is open.
4. A_{lens}/Σ_0 **degeneracy.** A proper MGCAMB run with (μ_0, Σ_0) as primary MG parameters will disentangle the partial degeneracy in the bobyqa run.

9 Assessment

EFC has progressed from a phenomenological alternative to a minimal, testable extension of GR. Two results were established numerically for the first time in this session.

First, K_0 converges to the kill-test sweet spot from BAO and lensing data alone, without high- ℓ CMB. This is independent confirmation of the μ -signal.

Second, the parameter shift at EFC minimum ($H_0 \uparrow$, $\Omega_m \downarrow$, $\sigma_8 \downarrow$) is the exact pattern predicted analytically from $\mu < 1$ dynamics. The optimizer followed the physics.

What EFC is not: proven correct; decisively better than Λ CDM by Bayesian standards; tested with a full posterior.

What EFC is: a real competitor to Λ CDM that cannot be rejected on current data, with $\Delta\chi^2 \leq 0$ across all tested dataset combinations, and whose parameter shifts point toward known cosmological tensions as a consequence of the theory, not a fit target.

Remaining test:

cobaya-run efc_bridge_full.yaml (plik_lite + lowl + lensing + BAO + Pantheon+, full MCMC)

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